

The Hand That Gives

Pre-PIE *de-* and the Polyfunctional Architecture: Body-Part Noun, Arm-Word, and Heteroclititic Gift-Noun on a Single Pre-PIE Element

Paper 1 of the der- project

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Abstract

A comparative cluster first proposed by Marstrand in 1919 — Hittite *dār-na-áš* (a measure-noun derived from the Akkadogram UP.NI "fist"), Old Irish *dorn*, Welsh *dwrn*, Gaulish *durno-*, Old Irish *derna* "palm," Greek δῶρον "palm of hand; gift," and Greek δάρις (Hesychius) — was incorporated into Pokorny's *IEW* (1959: 203) and remained in the standard handbook for half a century. Schrijver (1995) labeled the δωρ-/δαρ- relation "obscure"; Beekes (2010) excluded the Celtic forms from δῶρον on grounds of u-vocalism; Matasović (2009) classified Celtic **durno-* as probably non-IE; Demiraj (1990, 1997) reversed the Albanian *dorë* connection on a Tagliavini citation that does not deliver its claimed exclusion. The present paper restores Marstrand's analysis on the formal foundation that was unavailable to him: Clayton's (2023) reconstruction of the **-wṛ-/-(e/o)n-* heteroclitite class, with which Blažek (1999) had independently begun to formalize the same cognate set, plus a decomposition of **de-* as a free pre-PIE morpheme productive across multiple grammaticalization pathways.

Under this reading, the cluster reflects three derivational outputs on a single pre-PIE element **de-* "open hand": a locative-extended body-part noun **de-r-* (yielding δάρις, *derna*, Continental Celtic *DERNV*[], Albanian *dorë*); a heteroclititic gift/measure-noun **deh₃-wṛ-/-(e/o)n-* (yielding δῶρον, Latin *dōnum*, Celtic **durno-*, Latvian *dūre*, Anatolian *tarawar*); and an arm-noun **dous-* (Pokorny *IEW* 226), reconstructed as **de-* + **(o)us-*, with Sanskrit compound stem *dor-* attested since the Śatapatha Brāhmaṇa, plus Avestan *daoš-*, New Persian *dōš*, Old Irish *doē*, Latvian *pa-duse*, and Slovene *pâzduha*. The Celtic *u* that frustrated Beekes is the regular reflex of the Proto-Nuclear-IE metathesis **wṛ > *ru > *ur*. The δωρ-/δαρ- alternation is the strong/weak stem contrast of a single heteroclititic paradigm. The Albanian *dorë* connection rests on a 117-year published lineage running Pedersen 1909 → La Piana 1937 → Belardi 1948 → Pokorny 1959 → Schindler 1967 → Frisk → Beekes 2010 (preserving the Albanian connection while severing the Celtic) → Vermeer 2008 → DGE, where Vermeer's recent reconstruction **ǵ^hēr-* without

sibilant is structurally identical to the framework's **dē-r-* except for the initial consonant.

The framework adopts the **coexistence reading**: PIE **ǵhesr-/or-/to-* (Greek χεῖρ, Hittite *kessar*, Tocharian *tsar/ṣar*, Armenian *jeṛn*; Sanskrit *hāsta-*, Avestan *zasta-*, Old Persian *dasta-*) is accepted as the source of the cross-IE "hand-as-joint/grasp" word, with the framework claiming only that the **der-/deh₃-r-/dous-* cluster belongs to a separate PIE etymon centered on "open palm, hand-as-measure, arm-as-extension." The coexistence is methodologically aligned with Pokorny's own taxonomy, which kept **der-* (IEW 203) and **ǵhesor-/ǵhesr-* (IEW 447) as separate entries.

Probability conditional on the framework's central decomposition: 40–50%.

1. Introduction

The comparative cluster under discussion has been recognized as a unit since Marstrand assembled it in 1919, within four years of Hittite's decipherment by Hrozný. On page 128 of *Caractère indo-européen de la langue hittite*, in the course of arguing that Hittite *a* in *dār-na-áš* descends from IE **o*, Marstrand identified the Hittite measure-noun (alternating with the Akkadogram UP.NI "fist, hand, handful") with Old Irish *dorn*, Welsh *dwrn*, and Gaulish *durno-*, and in footnote 1 he extended the comparison to Old Irish *derna* "palm" (in apophonic alternation), to Greek δῶρον (with the Hesychian gloss παλαιστή "palm-width"), and to Greek δάρις (with the Hesychian gloss σπιθαμήν "span"). The cluster is presented with explicit ablaut analysis, with the Hesychian apparatus, and as evidence for a sound law: Marstrand offered it not as an etymological aside but as structural support for his phonological argument about Hittite vocalism.

Pokorny incorporated Marstrand's cluster into IEW 203 in 1959, classifying the forms under **der-* "Spanne der Hand," and the comparison remained the standard handbook treatment for the next half-century. Three late-twentieth-century departures from this lineage motivate the present paper. Schrijver (1995: 58) characterized the δῶρ-/δαρ- relation as "obscure." Beekes (2010), in the *Etymological Dictionary of Greek*, separated the "gift" and "palm-breadth" senses of δῶρον on the grounds that Celtic "points to PIE **u*, which excludes any relationship with δῶρον." Matasović (2009) classified Celtic **durno-* as probably non-IE in origin in the absence of a viable IE etymology for the *u*-vocalism. A fourth departure, in the Albanological tradition, is Demiraj's (1990, 1997) reversal of the Pokorny-La Piana attribution of Albanian *dorë* to the cluster, on the authority of a Tagliavini

citation that on examination does not deliver the claimed exclusion.

The framework restores Marstrander's analysis on a formal foundation that was unavailable to him. Clayton (2023), in a UCLA dissertation, systematically reconstructs a class of PIE neuter heteroclites with strong stems in **-wr̥-* and weak stems in **-w(e/o)n-*, and demonstrates that a regular metathesis **wr̥ > *ru* operated at Proto-Nuclear-IE depth — after the Anatolian split, before branch separation. Clayton independently identifies **deh₃-* as a root that forms heteroclites of this class. Blažek (1999), publishing in *Tocharian and Indo-European Studies* 8, had independently arrived at the same heteroclitic structure for the same cognate set, formalizing it as **dwor-eH₂-wr̥-/w(e)n-* "handfulness" and adding the Anatolian witness *tarawar* "measure of capacity" with the corresponding verb *tarāwi(ya)-* "to hand over, deliver." The cluster reflects two derivational outputs on a single pre-PIE element **de-* "open hand": the locative-extended body-part noun **de-r-* (Hesychian δάρις, Old Irish *derna*, Continental Celtic *DERNV*) and the heteroclitic gift/measure-noun **deh₃-wr̥-/w(e/o)n-* (δῶρον, *dōnum*, Celtic **durno-*, Latvian *dūre*, Anatolian *tarawar*).

A third derivational sibling on the same root is recovered in PIE **dous-* "arm" (Pokorny IEW 226), reconstructed within the framework as **de- + *(o)us-*. The cross-branch attestation is robust — Sanskrit *dóṣ-* with compound stem *dor-* (attested since the Śatapatha Brāhmaṇa), Avestan *daoš-* "upper arm, shoulder," New Persian *dōš* "shoulder," Old Irish *doē* (gen. *doat*) "arm," Latvian *pa-duse* "armpit," Slovene *pâzduha*, Old Bulgarian *pazucha* "κόλπος" — and the *-s/-n* heterocclisis *dóṣ-/doṣán-* places the arm-noun in the same archaic body-part class as **akṣ-/akṣán-* "eye," **asth-/asthán-* "bone," and **ās-/āsán-* "mouth," which is also the class to which the framework's **de-r-* and **deh₃-wr̥-/w(e/o)n-* belong.

The framework recovers Albanian *dorë* "hand" as a member of the cluster, supported by a 117-year published lineage running Pedersen 1909 → La Piana 1937/1942 → Belardi 1948 → Pokorny 1959 → Schindler 1967 → Frisk → Beekes 2010 → Vermeer 2008 → DGE. Vermeer (2008) reconstructs **ǵ^hēr-* without sibilant, structurally identical to the framework's **dē-r-* except for the initial consonant. The framework's **dē-r-* uses only the **ē → o* sound change that all parties accept; the standard **ǵh-onset accounts require a *ǵ^h → d** shift that Tagliavini (1936) and Orel (1998) admit to be irregular.

The framework adopts the **coexistence reading**: PIE **ǵhesr-/or-/to-* (Greek χεῖρ, Hittite *kessar*, Tocharian *tsar/šar*, Armenian *jeṛn*; Sanskrit *hāsta-*, Avestan *zasta-*, Old Persian *dasta-*) is accepted as the source of the cross-IE "hand-as-joint/grasp" word, with the framework claiming only that the **der-/deh₃-r-/dous-* cluster belongs to a separate PIE etymon centered

on "open palm, hand-as-measure, arm-as-extension." The coexistence is methodologically aligned with Pokorny's own taxonomy, which kept **der-* (IEW 203) and **ġhesor-/*ġhesr-* (IEW 447) as separate entries with the latter explicitly question-marked and the **ġhesr- → dorë* derivation explicitly ruled out on phonological grounds.

The paper is organized as follows. §2 sets out the polyfunctional architecture of the four-paper project. §3 presents Marstrander's 1919 comparison. §4 develops the body-part noun **de-r-*. §5 develops the **dous-* "arm" branch. §6 develops the heteroclitlic gift-noun. §7 develops the Celtic **durno-* derivation. §8 develops the $\delta\omega\rho/\delta\alpha\rho$ - alternation and the Vitruvius testimony. §9 develops the Albanian *dorë* extension. §10 develops the Anatolian evidence and the chronological architecture. §11 engages the late-twentieth-century departures. §12 sets out scope and limitations including the coexistence reading. §13 sketches the remaining papers. §14 gives the probability assessment. §15 provides authorship and contributions.

2. The Polyfunctional Architecture

The framework rests on the reconstruction of a pre-PIE element **de-* "hand" that was productive across multiple derivational pathways simultaneously. Body-part terms are cross-linguistically among the most fertile sources of grammaticalization and morphological productivity: the hand reaches, points, conveys, spans, and gives, and these functions leave distinguishable lexical and grammatical residues. Heine and Kuteva (2002) document the resulting range of pathways — HAND > LOCATIVE, HAND > AGENT MARKER, HAND > GIVE, HAND > DIRECTIONAL — across typologically diverse languages. The proposal is that pre-PIE instantiated several of these pathways in parallel, and that the comparative evidence in IE reflects the different outputs.

The full architecture of the four-paper project is shown in Table 1. The present paper develops the first three rows; Papers 2–4 develop the remaining branches. The reconstruction does not introduce a new PIE root: the bare **de-* and the locative **-r-* are independently attested morphemes, and the framework is a proposal about their prior productive relationship.

Root + suffix	Surface form	Meaning	Paper
<i>*de- + *-r-</i>	<i>*de-r-</i>	body-part noun ($\delta\acute{\alpha}\rho\iota\varsigma$, <i>derma</i> , <i>DERNV</i> [], Albanian <i>dorë</i>)	1
<i>*de- + *-h₃-wr̥-/w(e/o)n-</i>	<i>*deh₃-wr̥-</i>	gift/measure-noun ($\delta\acute{\omega}\rho\upsilon\nu$, <i>dōnum</i> , <i>durno-</i> , <i>tarawar</i>)	1
<i>*de- + *-(o)us-</i>	<i>*dous-</i>	arm-noun (Skt. <i>doṣ-/dor-</i> , Av. <i>daoš-</i> , OIr. <i>doē</i>)	1

<i>*de-</i> + <i>*-r-</i> + <i>*-o-</i>	<i>*(t)ero-</i>	bilateral-contrastive suffix	2
<i>*de-</i> + <i>*-u-</i> + <i>*-o/-i-</i>	<i>*d̥u-o-/d̥u-i-</i>	numeral 'two'	2
<i>*de-</i> + <i>*-h₃-</i>	<i>*deh₃-</i>	give-verb (post-Anatolian)	3
<i>*de-</i> + postposed <i>*-a</i>	<i>*de + a → *dā-</i>	'take' (Anatolian)	4

Table 1. Derivational pathways on pre-PIE **de-* across the four-paper project.

Two structural features of this skeleton bear on the central claims of the present paper. First, the locative **-r-* that appears in both the body-part noun (row 1) and the contrastive suffix (row 4) has independent support from three published treatments: Lipp (2009), reconstructing the PIE heteroclitlic r/n-stem paradigm, treats **-r* and **-én* as parallel locative endings; Bauhaus (2019) argues that **-r* is a true PIE locative case ending rather than an adverbial particle; and Kahl (2024), working on degree morphology, independently derives the secondary comparative of Indo-Iranian and Greek from "an archaic locative morpheme **-er*." The convergence of three independent treatments from three scholarly traditions is the strongest available warrant for treating **-r-* as a productive locative morpheme in early PIE.

Second, the bare **de-* directional particle is not a theoretical abstraction: it survives as the Greek allative clitic *-δε* (productive in Mycenaean by 1400–1200 BCE; Lejeune 1961, van Hes 2021), as Old Church Slavonic *do*, as Proto-Germanic **tō* (English *to*, German *zu*), and in Celtiberian *DV* (Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45). Paper 3 develops these as the grammaticalized descendants of a pre-PIE hand-element via the documented HAND > DIRECTIONAL pathway. Paper 4 develops the further claim that Hittite *dā-* "to take" preserves a fossilized univerbation **de + a* in which the bare hand-element combined with the postposed directional particle **-a* and contracted by hiatus.

The three rows of Table 1 that the present paper develops — **de-r-* (body-part), **deh₃-w̥r-/w(e/o)n-* (gift), **dous-* (arm) — sit in the same archaic body-part heteroclitic class as **ak̥s-/ak̥sán-* "eye," **asth-/asthán-* "bone," and **ās-/āsán-* "mouth." The cross-branch attestation of **dous-* (§5 below) and the Sanskrit compound stem *dor-* attested since the Śatapatha Brāhmaṇa show that the framework's preferred D-onset surface shape is anchored in the Vedic-prose stratum. The morphological structure **de-* + **-(o)us-* parallels the structure of PIE **deiu-o-* "god" (= **dei-* + **-u-* + **-o-*).

3. Marstrand 1919: The Original Comparison

Marstrand's comparison occupies page 128 of *Caractère indo-européen de la langue hittite*, in the section establishing that Hittite *a* descends from IE **o* in certain positions. The relevant passage reads:

Un exemple certain d'i.-e. **o* est présenté par l'hitt. *dàr-na-áš*, qui selon M. Hrozný (p. 41, note 2) Ks II 4 n° 4 alterne avec accad. UP.NI «poing, main, poignée». Je l'identifie avec le thème en **o* irl. *dorn*, gallois *dwrn*, gaulois *durno*- «poing, poignée, largeur d'un poing (mesure)».¹

Translation: "A certain example of IE **o* is presented by Hittite *dàr-na-áš*, which according to Hrozný alternates with Akkadian UP.NI 'fist, hand, handful.' I identify it with the **o*-stem Irish *dorn*, Welsh *dwrn*, Gaulish *durno*- 'fist, handful, breadth of a fist (measure).'"

Footnote 1 extends the cluster:

En rapport apophonique avec ce thème est celui en *n*: irl. *derna* «la palme». D'autres degrés vocaliques se trouvent dans δῶρον · παλαιστή, δάρις · σπιθαμήν (Hésyche).

Translation: "In apophonic relation with this theme is the one in *n*: Irish *derna* 'the palm.' Other vocalic grades are found in δῶρον · παλαιστή, δάρις · σπιθαμήν (Hesychius)."

Three features of Marstrand's analysis bear on the framework's reconstruction. First, the comparison is structural rather than lexical-isolated. Marstrand offers it as evidence for a sound law (Hittite *a* < IE **o*), with the cognate set doing structural work in his phonological argument. Second, the ablaut analysis is explicit. Marstrand distributes the cluster across distinct grades: the **o*-grade theme contains the Hittite measure-noun and the Celtic forms; an *n*-stem theme contains Old Irish *derna* "palm"; further unspecified vocalic grades contain Greek δῶρον and δάρις. The framework's **e*-grade **de-r-* (body-part noun) and **o*-grade gift-noun analysis is a development of Marstrand's original distribution, with the locative **-r-* (Lipp/Bauhaus/Kahl) and the heteroclitic **-wr̥-/w(e/o)n-* (Blažek/Clayton) supplying the morphological mechanisms that were unavailable in 1919. Third, the Hesychian apparatus is in place from the start. Marstrand cites Hesychius for both Greek forms, with measurement-related glosses: δῶρον is glossed παλαιστή (the Greek measure for palm-width, four finger-breadths), and δάρις is glossed σπιθαμήν (the standard Greek "span"). The Vitruvian connection between δῶρον "gift" and δῶρον "palm" (De architectura 2.3.3, on which §8 below) is corroborated by Hesychian lexicography that Marstrand already had in evidence.

Marstrand's *dàr-na-áš* corresponds to what the standard Hittite etymological dictionaries now treat as the measure-noun *tarna-* (Tischler HEG

III, Lieferung 9, p. 195), distinct from the verb *tarna-* "to release, let go." The framework's connection to the verbal "hand-over" semantics runs through the related Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" and the corresponding noun *tarawar* "measure of capacity" (Tischler 1991: 155–156; Melchert 1993: 211; Blažek 1999; Hawkins 2000: 259–261), discussed in §10 below.

4. The Body-Part Noun **de-r-*

The framework's **e*-grade analysis treats Marstrander's *n*-stem *derna* as the **e*-grade output of a locative-extended hand-noun **de-r-*. The locative status of **-r-* rests on three independent published treatments.

Lipp (2009), reconstructing the PIE heteroclitic *r/n*-stem paradigm, argues that the locative singular forms of acrostatic heteroclitite neuters are built from root nouns combined with an adverbial particle: **-r* and **-én* as parallel locative endings meaning "at/in the place of (X)." On Lipp's analysis, the spatial semantics is primary and the body-part association is downstream.

Bauhaus (2019) advances a stronger version: **-r* is not an adverbial particle but a true PIE locative case ending — "a real case ending, not an Adverbialendung" — active in pronominal and nominal formations across the family: **k^wor* (Gothic *þor* "where"), **tor* (Gothic *þar* "there"), nominal locatives such as **ǵ^hǵm-ér* "on the earth."

Kahl (2024), studying degree morphology in a Harvard dissertation under Jeremy Rau, reaches the same conclusion from a third direction: "the origin of the 'secondary comparative' of Indo-Iranian and Greek is traced to an archaic locative morpheme **-er*." The three treatments are methodologically independent. The compositional reading of **de-r-* — "at the place of the hand-direction" — is the natural semantics given the attested body-part function of the forms.

The cluster's surface witnesses to the **e*-grade body-part noun:

Hesychian δάρις / δάρειρ, glossed σπιθαμή ("span of all fingers"). The form is an *i*-stem derivative of **de-r-*, morphologically transparent: the open palm as a locative body-part anchor, "the span at the place of the hand-direction." Marstrander 1919 already cited this gloss in his footnote 1.

Old Irish *derna* (Gaelic *deàrna*) "palm of the hand." Continues the unmarked feminine singulative **de-r-nā-*. Pedersen (1913, II: 110) places Old Irish *dernu* / gen. *dernand* "Handfläche" in the same paradigm as *gualu* / *gualand* "shoulder," *Ánu* / *Ánann* (the goddess), and *áru* / *árann* "kidney" — a Celtic-internal classification of *derna* alongside other paired body-part *n*-stems, six years before Marstrander published *Caractère indo-européen*.

Continental Celtic *DERNV* at the Bronze of Novallas (Stifter 2022) — a Celtiberian hospitium bronze that preserves the body-part noun alongside the directional particle *DV* (Paper 3's Celtiberian reflex of **de/do*). Two derivational branches of the bare hand-element appear in a single inscription, a convergence visible only through the framework's lens.

Sanskrit *dor-*, the compound stem of *doṣ-* "arm" (attested since the Śatapatha Brāhmaṇa: *dor-bāhavāṇi*; Mayrhofer EWAia I.749). The Sanskrit *dor-* is the surface shape of the framework's body-part noun on the same root that gives the arm-word **dous-*; treated in detail in §5 below.

Bauhaus's most structurally significant contribution for the broader framework is his reconstruction of a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots: **h₁upér* "on top" vs. **h₁upó* "to the top," **k(e)nter* vs. **knto*, **ndher* vs. **ndho*. This paradigmatic pair is the structural template the framework reconstructs on **de-*: locative **de-r-* (the body-part noun) versus directive **de + a* (Paper 4's analysis of Hittite *dā-*). Bauhaus's observation that the locative **-r* "ceased to be productive in all attested languages" while the directive member "could keep its productivity at least until Anatolian" directly supports the chronological asymmetry the framework reconstructs.

5. PIE **dous-* "Arm": The Third Body-Part Derivative

The framework includes PIE **dous-* "arm" (Pokorny IEW 226) as a third derivational sibling, reconstructed as **de-* + **(o)us-* on the same root that produces **de-r-* and **deh₃-wr̥-/w(e/o)n-*. The **(o)us-* extension places the arm-word in the same archaic body-part heteroclite class as **ak̥s-/ak̥śán-* "eye," **asth-/asthán-* "bone," and **ās-/āsán-* "mouth."

5.1 Pokorny IEW 226 directly

Pokorny's **dous-* entry occupies five lines on IEW p. 226. The entry, transcribed verbatim:

***dous-* 'Arm'.**

Ai. *dóṣ-* n. (m.), Gen. *doṣnáh* 'Vorderarm, Arm, unterer Teil des Vorderfußes bei Tieren', av. *daoš-* m. 'Oberarm, Schulter', npers. *dōš* 'Schulter'; air. *doē* (**dous-nt-s*), Gen. *doat* 'Arm'; lett. *pa-duse* (tiefstufig) 'Achselhöhle; Busen des Kleides'; sloven. *pâzduha*, *pâzdiha* neben *pâzuha*, *pâziha* 'Achselhöhle', und mit demselben d-Verlust (ein Erklärungsversuch bei Berneker 233f.) abg. usw. *pazucha* 'κόλπος'.

WP. I 782, Trautmann 64.

Pokorny's reconstruction stops at **dous-*. The entry contains no further internal etymology, no derivation from a **ǵh-onset root*, no claim of irregular sound change, and no cross-reference to **ǵhesr-/or-/to-* (which has its own separate entry on IEW p. 447). The cross-branch attestation is robust: Sanskrit *dóṣ-* (gen. *doṣṇáh*); Avestan *daoš-* "upper arm, shoulder"; New Persian *dōš* "shoulder"; Old Irish *doē* with genitive *doat* (from **dous-ŋt-s*); Latvian *pa-duse* (zero-grade, "armpit, bosom of the garment"); Slovene *pâzduha* and variants ("armpit"); Old Bulgarian *pazucha** "κόλπος".

5.2 Mayrhofer EWAia I.749 and the Sanskrit *dor-* compound stem

The post-Pokorny standard reference for Indo-Aryan etymology — Mayrhofer's *Etymologisches Wörterbuch des Altindoarischen* (1992: vol. I, p. 749) — confirms Pokorny's **dous-* reconstruction without further reduction. The relevant portion of Mayrhofer's entry:

dóṣ- n. Arm, Vorderarm (RV, ŚB [NomAkkSg *dóḥ*]; Kauś *doṣ-ī* Du, AVP + *dor-bhyām*, kl. *dor-bhiḥ* u.a.; ***dor°*** in Kompp. seit ŚB [*dor-bāhavāṇi*; dazu Hoffm, IJ 4 (1960) 30 = HoffmA 106]; neben *dóṣ-* vgl. *doṣán-* in AV + *doṣāṇ-ī* NomDu, ŚB *doṣṇ-áh* GenSg, u.a.: AiGr III 317f.); *doṣaṇi-yà-* im Arm befindlich (RV, AV)... - Iir., vgl. jav. ***daoš-*** Oberarm... mp. np. ***dōš*** Schulter... - **Idg. wohl **dous-*, vgl. air. *doē*, GenSg *doat* Arm, lett. *pa-duse*, sloven. *pâz-ducha* Achselhöhle, u.a. (Pok 226, KEWA II 69, Fraenkel 560af., 641a).**

Three observations are critical for the framework. First, the framework's preferred surface shape *dor-* is documented in Sanskrit compounds since the Śatapatha Brāhmaṇa, with attested forms *dor-bhyām* (instr.dat.abl. dual), *dor-bhiḥ* (instr. plural), and *dor-bāhavāṇi* (compound: "the arms"). The *dor-* form is not a late Sanskrit innovation; it is the ancient regular compound stem of the body-part noun, attested in the Vedic prose corpus from approximately 700–500 BCE. The standard Sanskrit phonological rule that produces *dor-* from nominative *doḥ* before a voiced segment is regular sandhi of the visarga; what matters for the framework is that the resulting *dor-* compound stem is the surface shape of the framework's body-part noun, attached to the meaning "arm," in Vedic prose attested from before 500 BCE.

Second, Mayrhofer reconstructs "Idg. wohl **dous-*" ("Indo-European probably **dous-*") with no further reduction. He cites Pokorny IEW p. 226 as his source, taking from that page only the **dous-* reconstruction itself.

Third, on the next page of his entry (EWAia I.750), Mayrhofer notes that the heteroclisis *dóṣ-/doṣán-* "wohl auf das Indoar. beschränkt, mag von anderen Körperteilnamen wie **akṣ-/akṣán-* (o. I 42f.), **asth-/asthán-* (I 150f.; s. auch I 181f. zu *ās-/āsán-*) bezogen sein" — possibly motivated by analogy with

other body-part heteroclites. This places **dous-* in the same archaic body-part heteroclite class that the framework identifies as the home of **de-r-* and **deh₃-wṛ-/-(e/o)n-*. The *dóṣ-/dośán-* alternation is an *-s/-n* heterocclisis, paralleling the *-r/-n* heterocclisis of **deh₃-wṛ-/-(e/o)n-*.

5.3 The phantom **ġheus-* and its absence in Pokorny

A widely-circulated online Pokorny mirror (proto_roots.en-academic.com) appends an editorial note to Pokorny's **dous-* entry deriving the **dous-* family from an "archaic root **ġheus*" via "a unique O.Pers., Balt., celt., Illyr.-Alb. phonetic mutation." This note is not Pokorny's. Direct examination of IEW p. 226 (transcribed verbatim in §5.1 above) shows that Pokorny's actual entry contains nothing of the kind: no **ġheus-*, no claim of "unique phonetic mutation," no derivation of the **dous-* family from a **ġh-onset root*. *Direct examination of IEW p. 447 (the **ġhesor-/*ġhesr-/*ġhesto-* entries) likewise contains no **ġheus-*, no claim of unification with **dous-*, and no claim of a "unique phonetic mutation." The **ġheus-* unification exists only on the web mirror.*

The empirical content of the mirror's "unique phonetic mutation" claim is also incorrect on its face. Old Persian *dasta-* "hand" (under Pokorny's **ġhesto-* entry) does derive from PIE **ġ^h* via a regular sound change **ġ^h → d* in Old Persian (paired with Sanskrit *h*, Avestan *z*); this is the standard correspondence for Indo-Iranian **ġ^h* in *OPers.* There is no "unique mutation" particular to the **dous-* family; Pokorny's **dous-* entry simply has no **ġ^h-** in it to begin with.

The framework's specific claim — that **dous-* itself decomposes as **de- + *(o)us-* on the root that gives **de-r-*, **deh₃-wṛ-/-(e/o)n-*, and the rest of the polyfunctional architecture — is a step beyond Pokorny's reconstruction (which treats **dous-* as primary). Mayrhofer, citing Pokorny, also treats **dous-* as primary. The framework's decomposition is a structural proposal not contradicted by either reference work; the reference works simply do not propose it.

5.4 The Hand as Tip of the Arm

The semantic relation between the body-part noun **de-r-* "open palm" and the arm-word **dous-* is the relation between the tip of an extension and the extension itself. The hand is the distinguished extreme of the arm; the arm is the long-narrow-rigid extension that bears the hand. Both belong to the same anatomical line and are commonly named on the same root in cross-linguistic data (Heine and Kuteva 2002: under HAND > ARM and ARM > HAND). The Sanskrit compound stem *dor-* surfaces on the meaning "arm" in Vedic prose; the Greek δάρις surfaces on the meaning "span (of the open hand)" in

Hesychius. The same root, distinguished by the morphological extension, partitions the anatomical line into its component members.

Persian preserves a small constellation of long-narrow rigid-extension words on neighboring roots — *dōš* "shoulder" (PIE **dous-*), *dār* "tree, wood" (PIE **deru-/dreu-*), *tir* "arrow, beam" — which suggests that the typological frame "long-narrow rigid extension with a distinguished tip" was productive in pre-PIE morphology. The further extension of this frame to spear-tips, walking sticks, and boundary markers is registered as future research in §13 below; the present paper restricts itself to the framework's anatomical core.

6. The Heteroclititic Gift-Noun **deh₃-wṛ_ṣ-/-w(e/o)n-*

The $\delta\omega\rho/\delta\alpha\rho$ - alternation that Schrijver labeled "obscure" and the Celtic *u* that Beekes treated as exclusionary can be addressed together, through Clayton's (2023) reconstruction of the PIE **-wṛ_ṣ-/-w(e/o)n-* heteroclitite class.

Clayton's dissertation systematically reconstructs a class of neuter heteroclitites with alternating strong stems in **-wṛ_ṣ-* and weak stems in **-w(e/o)n-*, and demonstrates that a regular metathesis **wṛ_ṣ > *ru* operated at Proto-Nuclear-IE depth — after the Anatolian split, before branch separation. Clayton independently identifies **deh₃-* as a root that forms heteroclitites of this class, with Sanskrit *dārú-* "liberal" (from **déh₃-wṛ_ṣ-*) and Prasun *pyōrū* "gift" (from **pro-deh₃-wṛ_ṣ-*) among his examples.

Blažek (1999), in *Tocharian and Indo-European Studies* 8, had independently arrived at a heteroclititic structure for the same cognate set, formalizing the cluster as **dwor-eH₂-wṛ_ṣ-/-w(e)n-* "handfulness" and including the Anatolian witness *tarawar*. Blažek's reconstruction differs from the framework's in one notational respect — he writes **dw-* with consonantal *w*, while the framework treats this as **de-* + **-u-* under Paper 2's decomposition of **dwo-* "two" — but the underlying morphological architecture is the same: a heteroclititic **-wṛ_ṣ-/-w(e)n-* formation on a **d(e)-r/-/(u)-* base, ablauting across the cluster.

With this framework in place, the derivation of $\delta\omega\rho\upsilon$ from the heteroclititic strong stem is direct. PNIE **deh₃-wṛ_ṣ-* undergoes the regular metathesis to give **deh₃-ru-*; the laryngeal **h₃* colors and lengthens the preceding vowel, giving **dō-ru-*; thematicization yields **dōr-o-n-*; Greek surfaces as $\delta\omega\rho\upsilon$. The standard exemplar of the **wṛ_ṣ > *ru* metathesis is **k^wetwṛ_ṣ-* "four" → **k^w(e)tru-*, reflected in Avestan *caθru-*, Gaulish *petru-*, and Latin *quadru-* (Clayton 2023: §2.2).

Lesbian Aeolic preserves the un-metathesized **k^wetwṛ_ṣ-* in Greek itself: $\pi\acute{\epsilon}\omicron\upsilon\rho\epsilon\varsigma$ in the compound ordinal $\pi\epsilon\omicron\upsilon\rho\epsilon-\kappa\alpha\iota\delta\acute{\epsilon}\kappa\omicron\tau\omicron\varsigma$ "fourteenth" (IG

12,2:82.3/4) and neuter πέουρα at Memnon Nr. 31.7 (Blümel 1982: 271–272). Thessalian and Boeotian show the parallel post-metathesis treatments (πέτταρες, ordinal πέταρτος, πετράγουνον). The dialect distribution within Greek confirms that the metathesis was variable across Greek branches at the relevant date, providing in-Greek precedent for the differential treatment of **k^wetw_r-* versus **deh₃-w_r-* that the analysis of δῶρον and δάρις requires. The mechanism is well-attested both cross-branch (Iranian, Celtic, Italic) and within Greek itself (Aeolic vs. the rest); its application to **deh₃-* follows from Clayton's independent identification of this root as a member of the relevant class.

The weak stem accounts for the remaining Greek and Latin forms. The heteroclitic weak stem **deh₃-w(o)n-* (zero-grade **dh₃-w(o)n-*) yields Greek δάνος "gift, loan" with regular zero-grade vocalization. Latin *dōnum* is the same weak stem, thematized, with the long *ō* from laryngeal coloring in the full grade.

Form	Language	Stem	Analysis
δῶρον	Greek	strong, metathesized	<i>*deh₃-w_r-</i> > <i>*dō-ru-</i>
δάνος	Greek	weak, zero-grade	<i>*dh₃-w(o)n-</i>
<i>dōnum</i>	Latin	weak, full grade, thematized	
<i>dārú-</i>	Sanskrit	strong, metathesized	Clayton (2023)
<i>durno-</i>	Celtic	strong, metathesized + revocalized	§7 below
<i>dūre</i>	Latvian	strong, Baltic development	Schrijver (1995)
<i>tarawar</i>	Hittite/Luwian	heteroclitic, full reconstruction	Blažek (1999); §10 below

Table 2. The heteroclitic cognate set under the framework's analysis, restoring Marstrand 1919's comparison.

7. Celtic **durno-* and the Metathesis Pathway

The Celtic derivation requires separate treatment because of the original *u* (the formal point that Beekes 2010 found exclusionary) and because the compound evidence bears on the base meaning.

The *u*-vocalism is attested in Continental Gaulish from the first century BCE — *Durnomagus*, *Durnacos*, *Dagodurnus* — predating any Brythonic sound change. The chain from PNIE **deh₃-wr̥-* to Celtic **durno-* runs as follows: the PNIE metathesis **wr̥* > **ru* gives **deh₃-ru-*; in non-final syllables adjacent to a labial **w*, the syllabic resonant **r̥* is *u*-colored, yielding **dur-* (the Lubotsky mechanism, documented for Greek and Indo-Iranian in Lubotsky 1994, 1997); a singulative suffix **-no-* then gives **durno-*, after loss of the laryngeal in pre-Celtic. The closest phonological analog in the wider literature is OKhotanese *durna-*, which Bailey (1961) traces from PIIr. **druna-* by the same metathesis and revocalization; Clayton (2023) cites this parallel for his Sanskrit data, and it is equally the best available analog for the Celtic development.

The compound evidence bears on the base meaning. Proto-Celtic **are-durnon* (Old Irish *aurdorn*, Welsh *arddwrn*, Breton *arzorn*) means "wrist"; **ambi-durnos* (Old Irish *imdorn*) means "sword hilt." Neither compound is semantically natural with "fist" as the base; both are natural with "hand" construed as an anatomical unit. Breton reinforces this reading synchronically: *dorn* forms the body-part dual *daouarn* (= *daou* "two" + *dorn*), a category Breton reserves for anatomically paired body parts (Ball & Müller 2009: 445–446). The derivation *dorn* → *dornad* "handful" (Pedersen 1913, II: 35–36, with the suffix marking container contents) likewise presupposes an open-hand reading of the base.

The semantic evidence from Celtic compounds and derivatives is consistent with a body-part noun meaning "hand" rather than a posture term meaning "fist," which aligns with Marstrander's distribution: the **o*-grade **durno-* covered the hand as an anatomical unit, with "fist" as the contextually salient configuration in martial and combative contexts. Old Irish and Welsh narrowed to "fist" because the martial context dominated in the Insular Celtic record; Breton, the Continental compounds, and Pedersen's pre-Marstrander glosses ("Hand" passim in vol. I §§184, 277) preserve the broader "hand" reading.

8. The δῶρ-/δάρι- Alternation and the Vitruvius Testimony

The alternation between δῶρον and δάρις — documented by Schrijver as formally obscure, recognized but unanalyzed by Marstrander — becomes transparent under the heteroclitc analysis: δῶρον is the thematized strong stem (**deh₃-wr̥-* > **dōru-*) and δάρις is an *i*-stem derivative of the bare

body-part noun (**de-r-*). They are not the same surface formation and need not exhibit the same vocalism; the connection is family relationship through the shared pre-PIE base.

Beekes's decision to separate the "gift" and "palm-breadth" senses of δῶρον into unrelated entries addresses a real formal problem within the standard reconstruction. The framework offers an alternative under which both senses descend from the same hand-root, one through the gift-noun branch of the heteroclite and one by family association with the body-part noun. That the two senses were understood as connected in antiquity is confirmed by Vitruvius, who states explicitly that the Greeks call the palm δῶρον because giving (*datio*) is called δῶρον in Greek, and a gift is always conveyed through the palm of the hand (*De architectura* 2.3.3).

Vitruvius's testimony does not settle the etymological question — an ancient author's folk etymology is not a formal derivation — but it is evidence that the semantic connection between the two senses was transparent to Greek speakers. Marstrander's Hesychian glosses point in the same direction: δῶρον · παλαιστή connects the gift-noun to the palm-width measure that Vitruvius describes; δάρις · σπιθαμήν connects the body-part i-stem to the standard Greek "span." The lexicographic and architectural traditions converge on a connection between the two senses that the framework explains as etymological family relationship rather than accidental polysemy.

9. Albanian *dorë*: The Framework's Principal Extension

The framework recovers Albanian *dorë* "hand" as a member of the comparative family. The recovery rests on a published lineage spanning 117 years that Demiraj (1990, 1997) reversed on grounds that, on direct examination of the primary sources cited, do not survive scrutiny.

9.1 The published lineage (1909–2010)

The Albanian vowel correspondences governing the *dorë* derivation are tabulated by Pedersen in 1909 (*Vergleichende Grammatik der keltischen Sprachen*, Bd. I, §32, pp. 47–48): "Idg. **ā* ... alb. *o* ... und idg. **ō* ... alb. *e*." Pedersen lists *dōnum* / δίδωμι / δῶρον / *dán* as the standard cluster on which the correspondence operates. This is the pre-La Piana published anchor for the phonological argument: PIE **ē* → *o* in Albanian is regular and was on record in the standard reference grammar of Celtic in 1909.

The connection of Albanian *dorë* with Greek δῶρον was first proposed by Gustav Meyer in 1883 (*Albanesische Studien* I, p. 92). Marco La Piana developed the phonological argument in two publications: a 1937 monograph *Intorno al riflesso della vocale "o" lunga dall'indoeuropeo e dal latino*

nell'albanese (Palermo) and a 1942 short note in *Indogermanische Forschungen* 58, p. 98. Walter Belardi's 1948 article in *Rivista degli Studi Orientali* 23, pp. 69–81 ("Ittito *ki-eš-šar*, greco χείρ") argued on independent grounds that the standard reconstruction **ǵ^hesr-* "hand" cannot yield Albanian *dorë* — the *-sr-* cluster would give Albanian **dostrë* by regular sound change, with the Albanian parallels *shtërpinj* and Liburnian *Εἴστροις* supplying the comparative basis. Pokorny propagated Belardi's argument in *IEW* 447, eleven years later, listing the explicit phonological output (**dostrë*) and concluding that Albanian *dorë* requires a non-s form. On p. 203, Pokorny placed Albanian *dorë* under **der-* "Spanne der Hand" alongside Greek δάρις and δῶρον, citing La Piana 1942 as the supporting reference. Schindler (1967, "Tocharische Miszellen," *IF* 72: 244) supported Pokorny's doubts about **ǵ^hesr-*.

The connection survives in three modern standard Greek etymological dictionaries. Frisk's *Griechisches Etymologisches Wörterbuch* (1960–72) preserves La Piana's **dōrom* > Albanian *dorë* derivation in the entry for δῶρον "Handbreite." Beekes's *Etymological Dictionary of Greek* (2010: 363) cites La Piana for the same connection — retaining the Albanian connection while elsewhere in the same entry separating the gift and palm-breadth senses on grounds of the Celtic *u*-vocalism (cf. §1 above). The *Diccionario Griego-Español* (DGE, online) reconstructs the body-part **deh₃r-* with explicit laryngeal **h₃* — essentially identifying the framework's **de-h₃-r-* in different notation.

The lineage spans 117 years and eight major Indo-Europeanists or standard reference works: Pedersen 1909 → Meyer (with reservation; cf. §9.2 below) → La Piana 1937 → La Piana 1942 → Belardi 1948 → Pokorny 1959 → Schindler 1967 → Frisk 1960–72 → Beekes 2010 → DGE.

Two further Albanological scholars deserve mention as critics rather than supporters of the standard **ǵ^hesr-* analysis. Stuart Mann, whose 1952 *Language* article (vol. 28, pp. 31–40) is cited in Demiraj's bibliography in support of **ǵ^herds*, in his 1948 *Historical Albanian-English Dictionary* explicitly criticized the standard treatment of Albanian **ǵh*-onset words: "Meyer's recognition and interpretation of the Indo-European elements of Albanian are at once both naive and inconsistent. It is difficult to see, for instance, how Indo-European palatal *k* can produce under similar conditions both *s...* and *th...*, or still worse, how palatal *g/gh* can produce at once *d* in *dorë*, *dimën*, *derr*, *dh* in *dhjes*, *dhëmb* and *dhallë*, and *z* in *zorrë*, *zot*, *zjarr* and *zë*." Mann 1952's competing reconstruction **ǵ^herds-* (with medial *-d-*) is not an endorsement of the standard **ǵ^hesr-* but an attempt to escape its phonological incoherence. Within Demiraj's 14-name bibliography in support

of *^ǵhesr- → dorë*, at least three (Meyer in part, Mann 1948 + 1952, Schindler 1967) are critics or competitors rather than substantive supporters.

9.2 Pokorny IEW 447 directly: doubt at the source

Demiraj's *AE* p. 140 cites Pokorny IEW p. 447 as the source he then overturns. Direct examination of IEW p. 447 shows that Pokorny's text does substantially more than the cited handbook view acknowledges. The relevant entry, transcribed verbatim:

1. *^ǵhesor-, ^ǵhesr- 'Hand'?

Arm. *jeṛn* (*^ǵhes-r-*u*), Pl. *jeṛ-k'* 'Hand' (Meillet Esquisse 83); gr. *χείρ* f. 'Hand', Gen. *χειρός*, dor. *χηρός*, Dat. Pl. *χεροί* (danach die Form *χερ-*); toch. A *tsar*, B *šar* 'Hand' (nach Pedersen Tochar. 236 aus **kesar* =) hitt. *ki-eš-šar* (*kessar*) n. und *ki-es-si-ra-aš* (*kesseras*) m., f., Dat. Sg. *ki-iš-ša-ri* (*kesri* = gr. *χείρι*?); über lat. *ir*, *hir* 'hohle Hand' (aus osk.-umbr. **hēr* < **ǵhēsra*?) s. WH. I 649.

2. *^ǵhesto- 'Hand, Arm'.

Ai. *hásta-ḥ* m. 'Hand', av. *zasta-*, apers. *dasta-* ds.; lat. *praestō* 'zur Hand' aus **prae-hestod* (EM 805 f.)?; anders s. unter *stā-*; lit. *pa-žastē*, *pa-žastis* f. 'Raum unterm Arm, Achselhöhle'.

Da arm. *jeṛn* auch auf **ǵher-u* zurückgehen könnte und ebenso die gr. Formen besser aus **ǵher-s* erklärt werden können, da ferner **alb. *dorë* f. 'Hand', Pl. konson. *duar* eine Grundform **ǵhēr-* erfordert (**ǵhēs-* hätte **dostrë* ergeben)**, wäre mit Belardi (Riv. Studi Orient. 23, 69 ff.) zu erwägen, ob nicht fürs Hitt. und Toch. Metathese von -rs- zu -sr- anzunehmen sei, und die Gruppe 1 von 2 zu trennen und zu **ǵher-* 'greifen' zu stellen sei.

Three substantive features of this entry are decisive for the framework's restoration.

First, the question mark on the gloss. Pokorny titles the entry "**^ǵhesor-, ^ǵhesr- 'Hand'?*" — with a question mark. He is flagging the very meaning of the etymon as uncertain. Handbook compilers use this typographical signal precisely when the etymon they are describing may not actually have the meaning conventional treatments assign to it.

Second, the Belardi argument in Pokorny's primary text. The boldfaced clause translates as: "Albanian *dorë* f. 'hand', consonantal plural *duar*, requires a base form **ǵhēr-* (**ǵhēs-* would have yielded **dostrë*).". This is the Belardi 1948 argument in Pokorny's own voice: **ǵhesr-* cannot have produced Albanian *dorë*, because the regular Albanian outcome of **ǵhēs-* would have been **dostrë*. The standard handbook view that Demiraj defends propagates a

derivation that Pokorny — the very source it cites — explicitly held to be phonologically impossible for Albanian, on the same page Demiraj cites him from.

Third, Pokorny's tentative alternative is **ĝher-* "greifen," not **ĝheus-*. The semantic field of **ĝher-* "greifen" (IEW p. 442-443) is enclosed places — Greek *χόρτος* "enclosed place, courtyard," *χορός* "dancing-place" (originally "enclosed place"), *χόριον* "afterbirth," Latin *cohors*, *hortus* "garden as enclosed place," Old Irish *gort*, Welsh *garth*, Gothic *garda* "Hürde, Viehhof." The body-part word that comes from this root is *χόριον* "afterbirth" (the membrane that *encloses* the fetus), not *χείρ*. Pokorny's tentative reassignment moves Greek *χείρ* from **ĝhesr-* (where it sits problematically) to **ĝher-* "greifen" (where the receiving root has no other "hand" reflex to motivate the move and no escape from the same Albanian sound-law problem, since **ĝ^h → d* would be no more regular for **ĝher-* than for **ĝhesr-*).

The Albanian sound-law problem is therefore unsolved by either of Pokorny's two reconstructions on IEW p. 447. The framework's **de-r-* (with PIE **d* giving Albanian *d* by regular sound change) is the only proposal among the published candidates that does not require positing the irregular **ĝ^h → d* shift in Albanian.

9.3 The Demiraj reversal: anatomy of a phantom citation

Demiraj's *Albanische Etymologien* (1997: 140) reverses the published lineage. The relevant paragraph reads:

"Altererbtes Wort für 'Hand', das Bopp (Alb. 492) mit gr. *χείρ* 'Hand' verglich; hierher auch heth. *kessar*, toch. A *tsar*, B *šar* 'ds.' (s. auch Meyer EWA. **ĝ^hēsrā*; Pedersen Eberts RL I 225; Jokl IF 36 [1916/17] 132; Cimochoowski LP 2 [1950] 230; Mann Lg. 28 [1952] 34: **ĝ^herds*; Çabej SE III mit Literatur; Hamp, Posit. of Alb., Anc. I.-E. Dial. [1966] 115; Ölberg Untersuchungen 116; Huld BAE 54; Kortlandt, Arm. a. Alb., Mém. Acad. Roy. Belg. (Cl. Lettres) III [1986] 40). Der bisher ausschließlich angenommene Ansatz einer dehnstufigen Bildung — Vertreter wird von Pokorny (447: Grf. **ĝ^hes(o)r-* 'Hand'; s. auch Schindler IF 72 [1967] 244) in Zweifel gezogen, **der offenbar nur aus diesem Grund den Zusammenhang Meyers (AS I 92) mit gr. δῶπον 'Handfläche' vorzieht (zustimmend La Piana Intorno 29ff.; Pok. 203: **der-* 'Handfläche'). Diese Deutung ist mit Tagliavini (AD 101) auszuschließen.**"

Demiraj's argumentative move occurs in three steps within the boldfaced sentences. (1) Demiraj acknowledges that Pokorny doubted the **ĝhesr-* → *dorë* reconstruction. (2) Demiraj acknowledges that Pokorny preferred the

framework's etymology — Meyer's connection of *dorë* to Greek δῶρον "palm of hand" (with La Piana agreeing; with the entry sitting on Pokorny IEW p. 203 under **der-* "Handfläche"). (3) Demiraj overturns Pokorny's preference with a single citation: Tagliavini *L'albanese di Dalmazia*, p. 101 — "Diese Deutung ist mit Tagliavini (AD 101) auszuschließen" ("This interpretation is to be excluded with Tagliavini (AD 101)").

Direct examination of Tagliavini's *L'albanese di Dalmazia* (Olschki, Florence 1936) page 101 shows that the page contains entries for *dëkum* "morto," *dëlme* "pecora," *dër* preposition "fino, mentre," *dër (-a)* "porta," *dërdh* verb "versare, spandere," and *dërdhum* "diarrea, erpete." The page does not contain any discussion of Albanian *dorë* "hand," any mention of Greek δῶρον, any mention of **der-* "Handfläche," or any rejection of any alternative etymology for *dorë*. Tagliavini's actual *dor* entry, on AD 106, contains six lines citing Meyer, Brugmann, Walde-Pokorny, Pedersen, and Thumb, and never engages the **der-/δῶρον* alternative. The Demiraj rejection thus rests on a citation that does not deliver the rejection.

The phantom citation propagates further. Orel's 1998 *Albanian Etymological Dictionary* (p. 99), the post-Demiraj standard reference, also cites "TAGLIAVINI Dalmazia 101" in its bibliography for the *dorë* entry, repeating the empty reference. The chain Tagliavini AD 101 → Demiraj AE 140 → Orel 1998: 99 carries the same phantom forward in time.

Tagliavini's own 1936 inaugural lecture *La lingua albanese* (Padua, 6 February 1936; published in *Studi Albanesi* V-VI, 1935-36, pp. 5-33), at pp. 13-14, contains two further admissions that bear on the framework's case. First, on p. 13: "all'indoeuropeo **ǵ* (palatale sonora) e **ǵʰ* (palatale sonora aspirata) **l'albanese risponde con dh (che in determinati casi si muta in d)**" — Albanian responds to PIE **ǵ/ǵʰ* with *dh*, with *d* only "in certain cases," with no rule specifying the conditioning. Second, on p. 14: "**ē* viene reso in albanese con *o*, p. es. *lodh(ë)* «stancare» < **lēd-*" — PIE **ē* → *o* is regular in Albanian. Tagliavini's own published lecture documents that the **ǵʰ* → *d* step required by the standard view is irregular and that the **ē* → *o* step required by the framework is regular.

9.4 Hamp 1966: a tentative review passage, not a substantive defense

Hamp's "The Position of Albanian" (1966) is one of two pre-2000 pillars in Orel's 1998 *dorë* entry — the other being Pisani *Saggi* 121. Direct examination of the cited page shows what Hamp actually wrote:

 By way of orienting ourselves, summarizing open issues, and correcting some unevennesses in past scholarship, we will now consider matters dealt with in Porzig's *Gliederung*.

First, some corrections. *dorë* 'hand' (p. 187), χεῖρ, etc. are taken back to **gher-* 'greifen', i.e., **gher-s-*; of course, in the light of Hittite the pre-form is something like **ghěsr* (fem. in Albanian < old neut pl. ?). The apparent reconstructions for Albanian are sg. *dorë* < **ghē(s)ră*, pl. *duar* < **ghē(s)res* (C-stem). An alternative, preserving the gender considerations along with formal shape, is *dorë* fem. < (by form-class analogy) *neuter < **ghē(s)r-An* (in Jokl's symbols) = **ghē(s)r-om*, thematized from **ghěsr*; *duar* < **dōr* < **geră* < **ghē(s)ra*.

This is the entire treatment of Albanian *dorë* on p. 115. Six lines, in a section explicitly framed as review and correction of Porzig's *Gliederung*. Three features bear on the framework's case. First, Hamp uses tentative language throughout: "are taken back to" (passive, no agent), "is something like," the question mark on the gender derivation "(fem. in Albanian < old neut pl. ?)", "the apparent reconstructions," "an alternative." None of these is the language of substantive defense; all are the language of tentative review. Second, Hamp's "alternative" reconstruction posits two unattested intermediate stages (**geră*, **dōr*) to derive Albanian *duar* from PIE **ghē(s)ra*, with the **gh* > *g* > *d* shift happening across two reconstructed waypoints. Third, Hamp does not engage Pokorny IEW p. 203, Meyer AS I 92, or any of the framework's preferred alternatives; he is reviewing Porzig and registering uncertainty within Porzig's framework. The Hamp pillar in Orel 1998 thus turns out, on direct examination, to be a tentative review passage with explicit uncertainty markers, not a substantive original argument.

9.5 Huld 1984: internal hedging

Huld's *Basic Albanian Etymologies* (1984) p. 54 is one of the substantive Albanological treatments cited by both Demiraj and Orel. The relevant prose:

Dorë was related to Gk χεῖρ 'hand' by Meyer (1891b:72) cf Arm *jeŕn*, Hitt *kiššar* both also 'hand'. This account was accepted by Jokl (1923a:90) and Tagliavini (1937:101). Hamp (1966a:115) noted that the singular arose from a collective neuter plural, **ghěsrā*, a reconstruction also made by Pisani (1955:121). **Pokorny (1959:203) follows La Piana's derivation from **dōrom*; but Albanian *o* is from either **ee* or **aa*, never **oo* and this derivation was rightly dismissed by Tagliavini.**

PIE */*gheesr-*/; IEW 203 √*der-1* 'Spanne der Hand'.

Huld's argument against the framework is: (a) La Piana's reconstruction is **dōrom* (with **ō*); (b) Albanian *o* does not come from PIE **ō*; (c) therefore **dōrom* cannot give Albanian *dorë*. This argument is correct on its premises — Pokorny's preferred **dōrom* derivation does have a phonological problem —

but the framework's preferred reconstruction is not **dōrom*. The framework reconstructs **dē-r-eh₂-* (lengthened-grade feminine *r*-stem), an **ē*-grade form, not an **ō*-grade form. Per Huld's own statement, Albanian *o* is regularly from **ē* (his "**ee*"). The framework's **dē-r-eh₂-* → **dērā* → Albanian *dorë* uses the exact sound law Huld endorses: **ē* → Albanian *o*. Tagliavini's own 1936 lecture documented this same change with the example *lodh(ē)* "tire" < **lēd-*. The lengthened-grade reconstruction passes Huld's filter cleanly.

Huld also hedges by including IEW 203 $\sqrt{\text{der-1}}$ "Spanne der Hand" in his terminal references. He has dismissed La Piana's specific **dōrom* but cites Pokorny's **der-* entry as a residual reference. The framework's **dē-r-* sits exactly there, in Huld's own bibliographic conclusion.

Finally, Huld misreports Hamp's tentative passage as a settled finding: "Hamp (1966a:115) noted that the singular arose from a collective neuter plural, **ghēsra*." Hamp's actual passage (§9.4 above) used a question mark on the gender derivation. Huld drops the question mark and propagates a tentative speculation as a noted finding. The propagation pattern is the same as the Tagliavini citation: tentative claims becoming settled claims as they pass through subsequent handbooks.

9.6 Vermeer 2008: the standard view abandons the sibilant

Vermeer's "The Prehistory of the Albanian Vowel System" (2008) is the most recent specialist treatment of the relevant phonology. Two passages are decisive for the framework's case.

On p. 592, Vermeer lists four candidate sources of Albanian vowel length, with *dorë* placed in category 4: "Compensatory lengthening if a following **s* is lost in the position before a resonant. The standard example is *dorë* 'hand' (e.g. B. Demiraj 1997a: 44, 140; Orel 2000: 15), **which however has also been explained as an analogical lengthened grade (e.g. Huld 1984: 54, 15).**" Vermeer continues: "Although the jury is still out on (3) and (4), nothing in what follows depends on a decision about these cases." Within the standard view itself, the source of length in *dorë* is unresolved between Demiraj/Orel's compensatory-lengthening account and Huld's analogical-lengthened-grade account.

On p. 595, Vermeer gives his own reconstruction: "**ē* > *o* (= **ā*), e.g. *dorë* 'hand' < **ghēr-* (the source of the length of **ē* is controversial in this example, see section 2)." Note the absence of *-s-* in the reconstruction. Vermeer reconstructs **ghēr-*, not **ghēs-*. He has implicitly accepted Belardi's and Pokorny's argument that **ghēs-* would have given Albanian **dostrë*, and reconstructs a sibilant-less form to escape the problem. Vermeer's **ghēr-* is structurally identical to Pokorny's tentative **gher-* "greifen" alternative on

IEW p. 447 — and structurally identical to the framework's **dē-r-* except for the initial consonant.

Vermeer's footnote 3 on the previous page is the second decisive passage. Discussing the **ō > e* reflex (the rule Huld 1984 invokes to dismiss **dōrom*), Vermeer adds: "Despite complete consensus about this reflex I do not think there is a single example all interested scholars regard as reliable." The very sound law Huld uses to dismiss the framework's earlier-form ancestor is described by Vermeer as having no example all specialists regard as reliable.

The implications for the framework are favorable on three fronts. First, the Pisani-Hamp pillars in Orel 1998 are vacated. Orel's pre-2000 argument cited only Pisani *Saggi* 121 (the spirant-loss rule) and Hamp *Anc. IE Dial.* 115 for the **ǵ^hesr- → dorë* derivation. If Vermeer has moved the standard view to **ǵ^hēr-* without sibilant, Pisani's spirant-loss rule becomes inapplicable (no **s* to lose), and Hamp's **ǵ^hēsṛā* collective plural becomes a non-starter on the same grounds. Second, the remaining difference between standard view and framework reduces to one phoneme. Vermeer's **ǵ^hēr-* and the framework's **dē-r-* are structurally identical except for the initial consonant. Both have long **ē*, both have the **r*-stem, both rely on the **ē → o* sound change. The framework wins on parsimony: **d → d* is regular in Albanian; **ǵ^h → d* is irregular. Third, Huld's argument against **dōrom* is disarmed — the **ō > e* rule on which Huld's "rightful dismissal" of La Piana depends has, on Vermeer's own admission, no example "all interested scholars regard as reliable."

9.7 The framework's specific claim and convergence with the post-2000 standard view

Albanian *dorë* descends from a lengthened-grade body-part formation **dē-r-eh₂-* (lengthened-grade feminine *r*-stem on the framework's **de-r-*), with the Albanian *o < *ē* by the regular reflex documented in Pedersen 1909 §32, Tagliavini 1936: 14, La Piana 1937, Pokorny IEW 203, and Huld 1984: 54.

The framework uses only one Albanian sound change, **ē → o*, on which all parties agree. Tagliavini 1936 documents this as regular. Huld 1984 documents this as one of the two regular sources of Albanian *o*. Vermeer 2008 documents this as the regular reflex of **ē*. The framework's reconstruction is structurally identical to Vermeer 2008's **ǵ^hēr-* except for the initial consonant. The framework's specific claim is that the initial consonant should be **d-* rather than **ǵ^h-*, because **d → d* is regular in Albanian and **ǵ^h → d* is irregular. The argument is a parsimony argument about one phoneme.

The framework's IEW provenance is in Pokorny's own dictionary. The framework's **de-r-* "Spanne der Hand" is the analysis Pokorny preferred at

IEW p. 203, with La Piana 1942 cited there as the supporting reference. Demiraj's overturning of this analysis on the authority of Tagliavini AD 101 rests on a citation that does not deliver the exclusion. Huld's dismissal of **dōrom* on the **ō* > *e* sound-law applies to La Piana's specific reconstruction but not to the framework's lengthened-grade refinement, and the underlying sound law is, per Vermeer's own admission, not securely backed by reliable examples.

The framework's reconstruction is therefore the most parsimonious option in a field where the published consensus on Albanian *dorë* is not actually settled. Within the **ǵ^h-onset analysis itself, three competing accounts (Demiraj's compensatory lengthening; Huld's analogical lengthened grade; Vermeer's *ǵ^hēr- without sibilant) disagree on the phonological mechanism, and the post-2000 standard view (Vermeer 2008) has independently moved to a reconstruction that differs from the framework's by one phoneme. The framework's **dē-r-eh₂-* is the only reconstruction among the candidates that uses only the sound laws all parties accept. Combined with the IEW 203 √*der-* "Spanne der Hand" anchor in Pokorny's own dictionary, the published Pedersen-Meyer-La Piana-Belardi-Pokorny-Schindler lineage that survives in Frisk, Beekes, and DGE, and the documented phantom Tagliavini citation in Demiraj, the framework's case for Albanian *dorë** is defensible in print on standard methodological grounds, without overturning or replacing any sound law that the standard view accepts.*

10. Anatolian Evidence and Chronological Architecture

Marstrander's *dār-na-áš* corresponds to what Tischler's *Hethitisches Etymologisches Glossar* (1983, Lieferung 9) treats as the measure-noun *tarna* — a unit equal to half an *upnu* (Akkadian for "clenched fist," cf. Carruba StBoT 2, 1966, 11 Anm. 5), distinct from the Hittite verb *tarna-* "to release, let go." The framework's connection to verbal "hand-over" semantics runs through a related Hittite/Luwian etymon: *tarāwi(ya)-* "to hand over, deliver" (Tischler 1991: 155-156; Melchert 1993: 211) and the corresponding noun *tarawar* "measure of capacity" (originally "handful, fistful").

Hawkins (2000: 259-261) publishes the ISKENDERUN inscription, a Hieroglyphic Luwian dedicatory text on a tongue-shaped stone in the form of a ceremonial millstone, attributed to Laramas II of Gurgum (late 9th century BCE). The opening clause reads:

§ 1: (B) | wa/i-na *179.SCALPRUM-na ("*69")tara/i-wa/i-i-ha

"I provided(?) it (as) a millstone(?)"

The verb form *tara/i-wa/i-i-ha* represents */tarawiha/*, 1st singular preterite of the verbal stem *tarawi(ya)-* "provide," with the *-ha* ending characteristic of the Hittite *hi*-conjugation. The verb is preceded by logogram 69, which Hawkins (2000: 260) identifies as "a cursive, stylized form of 'hand'." Hawkins then opens an etymological discussion in which Oettinger (Stammbildung p. 379) groups four Hittite/Luwian forms as a single root family: Hittite *tarrawai-* "provide goods to temples" (KBo II.7); Hittite *tarawiya-* — Goetze defines this as a noun denoting a favorable condition, perhaps "endowment" (KBo XI.1); Cuneiform Luwian *tarami-* (debated); Hittite *tarrawa-** (KUB VI.15).

The HLuwian attestation is significant for the framework on three counts. First, it places the verbal "hand-over" semantics in the standard reference work for Hieroglyphic Luwian, written with a hand-pictogram. The HAND-source of the verb is preserved in the script. Second, Goetze's "endowment" reading of Hittite *tarawiya-* — an inherited noun on the verb stem — corroborates the framework's gift-noun analysis at the level of attested Hittite morphology, beyond what Blažek's reconstruction alone provided. Third, it grounds the framework's Anatolian witness in the standard reference works of Anatolian philology (Hawkins 2000, Oettinger 1979, Tischler 1991, Melchert 1993), with Blažek 1999 supplying the heteroclitc formalization rather than serving as the foundation.

Blažek's full reconstruction:

**dwor-eH₂-wṛ-/w(e)n-* "handfulness" > Anatolian **tarawar/-wan-* "handful" > "measure of capacity"; **dwōr-o-* > Greek δῶρον;
**dur-o-unt(s)/-went-* > Tocharian B *trau* "measure of capacity"; **dur-no-* > Celtic **durno-*; **dūr-iyā* > Latvian *dūre*.

The framework's reading aligns with Blažek's substantive reconstruction. The notational difference — Blažek's **dwor-* versus the framework's **de- + *-u- + *-r-* via the Paper 2 decomposition of **dwo-* "two" — does not affect the surface forms or the heteroclitc structure. The framework's specific contribution is the morphological decomposition that connects this cluster to the broader architecture of Papers 2-4: the **-u-* of **dwo-* is the same **-u-* that figures in the heteroclitc **-wṛ-/w(e/o)n-*, and the bilateral semantics of the numeral and the contrastive suffix **(t)ero-* are inherited from the same pre-PIE element.

Norbruis (2021: 243-249) argues, on Anatolian-internal grounds, that the original meaning of **deh₃-* was "to take" rather than "to give," with the give-meaning a post-Anatolian innovation through the semantic chain "to take" > (with goal/beneficiary) "to bring, convey, deliver" > (with animate recipient) "to hand over, to give." The intermediate stage — physical delivery of concrete objects to animate recipients — is exactly the meaning of

Hittite/Luwian *tarāwi(ya)*- "to hand over, deliver" (Tischler 1991; Hawkins 2000: 260). The Anatolian verb preserves Norbruis's reconstructed intermediate stage, in published form, on a heteroclite of the framework's reconstructed root family.

The chronological architecture is coherent. Body-part noun first (pre-Anatolian, **de-r-*); arm-noun second (pre-Anatolian, **dous-*, with Indo-Iranian preserving the Sanskrit compound stem *dor-* and Avestan/Persian preserving *daoš-/dōš*); directional particle third (pre-Anatolian, **de/do*); take-verb fourth (pre-Anatolian, Anatolian branch, **dā-*); give-verb and gift-noun fifth (post-Anatolian, PNIE, **deh₃-* and **deh₃-wr₆-*). The absence of the **deh₃-wr₆-* gift-noun from Anatolian as a verb of giving is expected: Anatolian preserves the root in *tarāwi(ya)*- "to hand over, deliver" at Norbruis's intermediate stage, with the abstract "gift" semantics post-Anatolian. Paper 4 develops the formal account of Hittite *dā-* "to take" as the Anatolian representative of the take-verb branch, deriving it from a pre-Anatolian directive univertation **de + a* that resolves the take/give polarity structurally. Pedersen (1913, II: 380) registers the earliest published recognition that the give-root has an *a*-allomorph that ordinary ablaut won't generate, citing the Continental Celtic 3sg perfect $\delta\epsilon\delta\epsilon$ of **deh₃-*; the *a*-shape that Pedersen plugged with Latin influence on the Celtic paradigm is the same shape Paper 4 resolves through the directive univertation.

11. Previous Accounts: The Late-Twentieth-Century Departures

With the published lineage in place, the framework's engagement with the standard view is best framed as a return to the pre-Beekes consensus rather than as a refutation of established scholarship. Three departures from the lineage warrant brief comment.

Schrijver's (1995: 58) "obscure" assessment accurately described the absence of a satisfactory formal account within standard PIE phonology. The framework does not so much correct Schrijver as supply the formal tool that was missing: the Clayton (2023) heteroclite framework was not available at the time of Schrijver's writing, and Blažek's (1999) parallel reconstruction had not yet appeared when Schrijver composed his treatment. Schrijver's parenthetical observation, citing Beekes, that "in the Greek forms, $\delta-$ might reflect **du-*" anticipates the morphological-identity argument developed in Paper 2.

Beekes's (2010) separation of the two senses of $\delta\omega\rho\omicron\nu$ and his exclusion of Celtic on grounds of *u*-vocalism are both formally motivated within the evidence available to him. Beekes wrote without access to Clayton 2023 and without reference to Blažek 1999 (which sits in a Tocharian-studies venue).

The framework offers an account under which the two senses are related and the Celtic *u* is the regular outcome of the **w_r > *ru* metathesis. The two accounts make different predictions about the morphological history and can be evaluated independently.

Matasović's (2009) substrate hypothesis followed from the absence of a viable IE etymology for Celtic **durno-*, which was the appropriate methodological response within the evidence available. The claim of the present paper is that a viable IE etymology is now available through the heteroclite framework. If that claim is correct, the substrate hypothesis is no longer necessary as an explanatory device.

The Demiraj reversal of the Albanian *dorë* lineage (1990, 1997) is engaged at length in §9 above; the citation chain on which it rests does not survive examination.

12. Scope and Limitations: The Coexistence Reading

Several clarifications are necessary to define the proposal's scope precisely.

The framework does not introduce a new PIE root. The bare **de-* and the locative **-r-* are independently attested morphemes; the proposal concerns their prior productive relationship. A reviewer committed to atomistic root reconstruction will resist the compositional analysis on principled grounds, and that resistance is coherent; the framework's claim is that the compositional reading is explanatorily superior to treating each reflex independently.

The framework adopts the **coexistence reading**. PIE **ǵhesr-/or-/to-* (Pokorny IEW 447) is accepted as the source of the cross-IE "hand-as-joint/grasp" word: Greek χεῖρ, Hittite *kessar*, Tocharian *tsar/šar*, Armenian *jeṛn*; via **ǵhesto-*, Sanskrit *hásta-*, Avestan *zasta-*, Old Persian *dasta-*. The framework does not contest this reconstruction. The framework claims only that the **der-/*deh₃-r-/*dous-* cluster (Greek δῶρον/δάρις, Celtic **durno-/derna*, Latvian *dūre*, Albanian *dorë*, Sanskrit *dor-/doṣ-*, Avestan *daoš-*, Old Irish *doē*) belongs to a separate PIE etymon centered on "open palm, hand-as-measure, arm-as-extension." This coexistence is methodologically aligned with Pokorny's own taxonomy, which kept **der-* (IEW 203) and **ǵhesor-/*ǵhesr-* (IEW 447) as separate entries with the latter explicitly question-marked. The two etyma are likely to have had specialized senses in early PIE: **ǵhesr-* the hand as joint/grasp; **de-r-* the open palm and hand-as-measure; **dous-* the arm-as-extension. Specialists who reject the **der-/*ǵhesr-* split can evaluate the framework's claims about the **der-* cluster independently of the question of whether **ǵhesr-* is also right for its own reflexes.

The reconstruction of **deh₃-* as the correct post-Anatolian verbal pre-form is not in dispute. The claim is about the internal morphological structure of that form — that **-h₃-* is a derivational extension suffix rather than an unanalyzable root consonant — and about the chronology of its addition. The phonological evidence for **h₃* in the post-Anatolian period is accepted; the question is only when and how the laryngeal was added.

The Marstrander 1919 attribution recovers the lexical comparison and the ablaut analysis for the published lineage; it does not absolve the framework of the need to defend the formal mechanism (the heteroclite structure, the metathesis pathway, the Lubotsky revocalization). Marstrander's specific phonological argument — that Hittite *a* in *dār-na-áš* descends from IE **o* — is accepted in Hittite historical phonology. The framework's reconstruction is consistent with that finding.

The older derivation of δῶρον from **deh₃-ro-* remains formally available within standard handbooks. The Clayton/Blažek heteroclite framework is adopted here because it accounts for the Celtic cognates that the older derivation could not, and because it brings the cluster into alignment with Anatolian *tarawar*. Specialists may reasonably prefer the older analysis pending further evaluation of Clayton's reconstruction.

The framework's **dous-* "arm" branch is registered with a probability lower than the gift-noun and body-part-noun branches. The reconstruction **de-* + **(o)us-* is a structural proposal beyond Pokorny's **dous-* (which Pokorny treats as primary); Mayrhofer treats **dous-* as primary as well. The framework's claim that **dous-* itself decomposes is a step beyond the standard view, even though the standard view does not contradict the decomposition (it simply does not propose it). The probability assessment in §14 reflects this.

Latvian *dūre* and the broader Baltic/Slavic "armpit"-derivatives are registered as supporting evidence but are not load-bearing; the Greek, Celtic, Anatolian, and Indo-Iranian evidence is sufficient to ground the lexical claim independently.

13. The Remaining Papers

Paper 2 ("A Better Hand") reconstructs the PIE contrastive suffix **(t)ero-* compositionally as **de-* + **-r-* + **-o-*, and develops the numeral **duo-/dwi-* "two" as a parallel derivational outcome on the same root. The bilateral character of **(t)ero-* — its consistent function of marking "one of a pair" across the family — is argued to be a morphological inheritance from the **-u-* element of pairedness, not a typological consequence of bilateral body symmetry. Dunkel's (2014) published etymology unifying the prefixal **ui-* and

the numeral **d̥ui-*, together with his reconstruction of the inherited compound **ui-tero-* (Vedic *vitarám*, OCS *vŭtoru* "the second"), provides the external support.

Paper 3 ("The Bare Hand") establishes **de-* as an independently attested PIE directional particle and decomposes the give-verb **deh₃-* as **de-* + verbal extension **-h₃-*. The strongest evidence for this bimorphemic decomposition is a class-by-class asymmetry in the Greek root-aorist 2pl imperative paradigm: roots the framework analyzes as bimorphemic (including **deh₃-* → *δότε*) preserve a bare-CV form in the 2pl, while roots the framework does not analyze as bimorphemic (including **steh₂-* → *στῆτε*) preserve the laryngeal-conditioned long vowel throughout.

Paper 4 ("The Hand That Takes") derives Hittite *dā-* "to take" from a pre-Anatolian directive univerbation **de + a*. The proposal fits a documented productive class in attested Hittite: the Set 1 directional system in which a local element is followed by the directional particle **-a* and contracts to a fixed adverb or preverb (*anda* < **en + a*, *katta* < **kmt + a*, *parā* < **pro + a*). The proposal resolves the take/give polarity — the fact that Anatolian's reflex of **deh₃-* means "take" while post-Anatolian reflexes mean "give" — as a structural consequence of the directional **a* "toward the speaker" combining with the hand-element. The Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" and the corresponding noun *tarawar* (§10 above) provide independent Anatolian witness to the framework's heteroclititic gift-noun branch at Norbruis's reconstructed intermediate stage.

The broader semantic frame — the "hand-as-tip" insight developed in §5.4 above, with its typological extension to spear-tips, walking sticks, boundary markers, and the Persian *dōš/dār/tir* constellation — is registered as future research. A possible Paper 5 would systematically develop the shape-primitive of "long-narrow rigid extension with a distinguished tip" across PIE and assess whether pre-PIE morphology productively encoded the primitive across multiple roots. A separate companion paper, currently in draft, develops a substrate divine-name complex **T/S/Š/T̥-V-R-M-* (Hermes / Turms / Sarruma) in the Anatolian-Aegean-Italian contact zone; the relation between that complex and the present framework's hand-root family is registered as future research and is not load-bearing for the present argument.

14. Probability Assessment

We assess the probability of the Paper 1 thesis — that the comparative cluster reflects derivational siblings on the same pre-PIE hand-element **de-* — at approximately 40–50%, conditional on its own evidence considered independently of Papers 2–4.

In favour of the analysis: the lexical comparison rests on a continuously-published lineage from Pedersen 1909 through Marstrander 1919 and Pokorny 1959 to Blažek 1999, Vermeer 2008, and the modern Greek etymological dictionaries (Frisk, Beekes 2010, DGE), with three independent twentieth-century departures (Schrijver, Beekes on Celtic, Matasović) each addressable on its own terms. The framework's formal mechanism — the heteroclitic **-wr̥-/w(e/o)n-* structure with regular **wr̥ > *ru* metathesis — has independent support in two published treatments (Clayton 2023, Blažek 1999) that arrive at the heteroclitic class from different empirical bases. The OKhotanese *durna-* parallel (Bailey 1961) is the closest phonological analog for the Celtic development in the literature. The unification of *δωρ-* and *δαρ-* as strong and weak stems of a single heteroclitic paradigm provides a structural account where Schrijver's label "obscure" has stood for thirty years. The Anatolian witness *tarawar/tarāwi(ya)-* (Hawkins 2000, Oettinger 1979, Tischler 1991, Melchert 1993, Blažek 1999) provides published Anatolian evidence at Norbruis's reconstructed semantic intermediate stage, anchored in the standard reference works of Anatolian philology. Lesbian Aeolic *πέουρες* (Blümel 1982: 271–272) supplies a Greek-internal pre-metathesis witness for **k^wetwr̥-*, the standard exemplar Clayton uses. The **dous-* witness (Pokorny IEW 226, Mayrhofer EWAia I.749) adds Indo-Iranian, Celtic, Baltic, and Slavic to the framework's branch coverage at the body-part level, with the Sanskrit compound stem *dor-* providing Vedic-prose surface attestation of the framework's preferred shape since the Śatapatha Brāhmaṇa. The Albanian *dorë* extension rests on a 117-year published lineage that the standard rejection (Demiraj 1990, 1997) does not adequately defend, and the post-2000 standard view (Vermeer 2008) has independently moved to a reconstruction (**ǵ^hēr-* without sibilant) that differs from the framework's by one phoneme.

Against: the compositional body-part noun **de-r-* is non-standard as a synchronic PIE reconstruction, even though its components (**de-* and **-r-*) are independently attested. The decomposition of **dous-* as **de-* + **-(o)us-* is a step beyond what Pokorny and Mayrhofer propose; standard reference works treat **dous-* as primary. The Celtic derivation involves more steps than earlier proposals, even if each step is independently motivated. The broader framework across all four papers requires a productive pre-PIE morphological system whose existence can only be evaluated by assessing the evidence together.

The probability at the level of the whole project — integrating the conditional probabilities across Papers 1–4 — is in the range 2–5%, with much of the evidence supporting multiple strata simultaneously rather than each stratum independently. We continue to regard the central claim as more likely

wrong than right at the marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

15. Authorship and Contributions

Chavis's contributions. The central hypothesis originates with Chavis: the identification of the dual attestation of Greek δῶρον — gift and palm-breadth — as evidence of a single etymological family was the project's founding insight. Chavis directed the research agenda at every stage: the hypothesis that **(t)ero-* descends from the same root; the identification of *ambidexter* as the morphologically transparent compound witness for Paper 2; the question "whence comes the *r* in **-tero-*?" that opened the compositional analysis of Paper 3; the hypothesis that Hittite *dā-* preserves a pre-Anatolian construction; and the decision to incorporate Clayton (2023) as the formal linchpin of the lexical argument. The decision to pursue the Albanian *dorë* extension despite Demiraj's published rejection, and to ground the framework's analysis in direct examination of the citation chain Demiraj invokes, is Chavis's. The decision to integrate the Marstrander 1919 attribution and the Anatolian *tarawar/tarāwi(ya)-* witness into Paper 1's central architecture — restructuring the paper around the published lineage rather than around a defensive engagement with the standard view — is Chavis's. The hand-as-tip semantic insight that frames the **dous-* "arm" addition; the Persian *dōš/dār/tir* constellation observation; the methodological pivot to the coexistence reading; and the systematic primary-source examination of the Albanian *dorë* citation chain (Demiraj p. 140, Pokorny IEW 447, Hamp 1966 p. 115, Huld 1984 p. 54, Vermeer 2008) are Chavis's. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude's contributions. Claude's contributions are primarily in source synthesis, exposition, and analytical development. Across the project, Claude located and read the relevant published literature: Marstrander 1919 *Caractère indo-européen de la langue hittite*, p. 128 + footnote 1, with the Hesychian apparatus; Blažek 1999 in *Tocharian and Indo-European Studies* 8, with the heteroclitic **dwor-eH₂-wr̥-/w(e)n-* reconstruction and the Anatolian *tarawar* witness; Pokorny IEW p. 226 directly (the **dous-* "Arm" entry); Mayrhofer's *Etymologisches Wörterbuch des Altindoarischen* (1992) vol. I p. 749 directly (the *dóṣ-* entry with the Sanskrit *dor-* compound stem dating since ŚB); Pokorny IEW p. 447 directly (the **ḡhesor-/*ḡhesr-/*ḡhesto-* entries with the question-marked gloss and the explicit **dostrë* rejection); Pokorny

IEW pp. 439–443 directly (the **ĝher*-/[**gher*- entries that bear on the alternative **ĝher*- "greifen" reconstruction); Demiraj AE p. 140 directly (the phantom Tagliavini citation); Hamp 1966 p. 115 directly (the tentative review passage); Huld 1984 p. 54 directly (the internally hedged entry); Vermeer 2008 pp. 592, 594–595 directly (the open-issue acknowledgment, the **ĝ^hēr*-without sibilant reconstruction, and footnote 3 on the unreliability of the **ō* > *e* sound law); Mann 1948's *Historical Albanian-English Dictionary* preface for the explicit critique of Meyer's **ĝh* treatment; Orel 1998: 99 directly for the structural diagnosis of the pre-◊ versus post-◊ bibliography organization; Tischler's *Hethitisches Etymologisches Glossar*, including the Marstrander 1919 attribution preserved at HEG III/9: 195 (Lieferung 9, 1983) for the *tarna*- measure-noun; Hawkins 2000 (Iron Age Hieroglyphic Luwian inscriptions) for the ISKENDERUN attestation of *tarawi*(*ya*)- with the 69 hand-pictogram and the Oettinger collation of four Hittite/Luwian forms; Blümel 1982 for the Lesbian *πέουρες* pre-metathesis witness; Pedersen 1909/1913 for the Albanian vowel correspondences in §32 and the Continental Celtic *δεδε* perfect in II:380; Clayton (2023) on the heteroclide class; Lipp (2009), Bauhaus (2019), and Kahl (2024) on the locative **-r*-; Norbruis (2021) on the semantic chronology of **deh*₃-; Stifter (2022) on the Bronze of Novallas; Lubotsky (1994, 1997) on the phonological mechanisms; Dunkel (2014) on the **ui*-tero- compound; Belardi 1948 on the Albanian **-sr*-impossibility; La Piana 1937 and 1942 (the latter via Frisk, Beekes, and DGE) on the **dōrom* derivation; the DGE laryngealized reconstruction **deh*₃*r*- that essentially identifies the framework's **de*-*h*₃-*r*-; Tagliavini L'albanese di Dalmazia directly, establishing that AD 101 does not contain the rejection Demiraj cites; Kloekhorst (2008, 2013) and Hoffner & Melchert (2008) on Hittite; Lejeune (1961) and van Hes (2021) on Mycenaean *-δε*; and ancillary sources throughout. Claude drafted the prose of all sections from Chavis's structural and substantive direction; identified Bauhaus's locative-directive paradigmatic pairs as published structural precedent for Papers 3–4; integrated the architectural alignment across all four papers at each revision stage; and proposed the structural integrations of the **dous*- "arm" branch, the expanded Albanian *dorë** engagement, and the coexistence-reading framing. The probability estimates in all four papers are Claude's own.

Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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